Srinivas Vivek Bokkisa

Postdoctoral Fellow | Geotechnical Engineering

School of Civil and Environmental Engineering, Georgia Institute of Technology | Atlanta, GA, USA

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Research Profile

Geotechnical engineer specializing in multiscale soil mechanics, constitutive modelling, and computational geomechanics. My research connects fabric-dependent soil behavior with mechanics-informed site characterization and performance assessment of geotechnical infrastructure.

Research areas: Advanced soil mechanics (fabric anisotropy, liquefaction, instability); computational geomechanics (MPM, FEM, DEM, constitutive integration); tailings and geo-infrastructure (performance-based assessments); site characterization and multiscale behavior (CPT, X-ray micro-CT).

Methods & tools: MPM, FEM, DEM; constitutive-model integration and calibration; X-ray micro-CT analysis; Python, Fortran, C++, MATLAB, Julia; PLAXIS, Abaqus, Anura3D, GeoTaichi, LIGGGHTS and Yade.

Academic Appointments & Professional Experience

Postdoctoral Fellow *May 2026–Present*
Georgia Institute of Technology, School of Civil and Environmental Engineering
- Investigate fabric-dependent soil behavior, constitutive modelling, and mechanics-informed characterization using computational geomechanics.
- Develop workflows for constitutive implementation and calibration, experimental-data interpretation, and boundary-value simulations.

Geotechnical Intern - Mining *Jun. 2024–Aug. 2024*
AECOM, Denver, CO, USA
- Performed large-deformation numerical analyses of a tailings-dam section to assess runout-material pressures on a proposed retaining-wall design using the Material Point Method.
- Developed a database from previous tailings investigations and integrated rapid-vane-shear data with co-located CPT data to evaluate engineering correlations for site characterization.

Education

Ph.D. in Civil Engineering *2026*
Georgia Institute of Technology
Dissertation: *Multiscale Assessment of Fabric in Granular Soils*; Advisor: Dr. Jorge Macedo; GPA: 3.91/4.0

M.S. in Computational Science and Engineering *2023*
Georgia Institute of Technology
GPA: 3.75/4.0

M.Phil. in Civil Engineering *2019*
Hong Kong University of Science and Technology
Dissertation: *Fabric Evolution in 3D Discrete Element Modelling of Soil Liquefaction*; Advisor: Dr. Gang Wang; GPA: 4.15/4.30

B.Tech. in Civil Engineering *2017*
Indian Institute of Technology Guwahati
GPA: 8.44/10

Peer-Reviewed Journal Articles

1. **Srinivas Vivek, B.**, Macedo, J., and Arduino, P. (2026). "On the integration of an ACST-based bounding surface model." *Computers and Geotechnics*. [DOI]
2. **Srinivas Vivek, B.**, and Macedo, J. (2025). "Challenges in NorSand to model CSD stress paths and proposed modifications." *Canadian Geotechnical Journal*. [Link]
3. **Srinivas Vivek, B.**, and Macedo, J. (2024). "Influence of anisotropic consolidation on the instability of loose granular soils under undrained and drained loading." *Canadian Geotechnical Journal*. [Link]
4. **Srinivas Vivek, B.**, Macedo, J., and Petalas, A. L. (2023). "Fabric anisotropy effects on static liquefaction under constant shear drained loading." *Computers and Geotechnics*. [Link]

Peer-Reviewed Journal Articles (continued)

5. **Srinivas Vivek, B.**, Macedo, J., Petalas, A. L., and Arson, C. (2022). "Assessing flow liquefaction triggering considering fabric anisotropy effects under the ACST framework." *Computers and Geotechnics*. [Link]
6. Kumar, A., and **Srinivas, B. V.** (2017). "Easy to use empirical correlations for liquefaction and no liquefaction conditions." *Geotechnical and Geological Engineering*, 35(4), 1383–1407. [Link]

Manuscript Under Review

1. **Srinivas Vivek, B.**, and Macedo, J. (2026). "Fabric Effects on Cone Penetration Testing: A Numerical Investigation." *Computers and Geotechnics*. Under review. [Link]

Conference Papers & Presentations

1. **Srinivas Vivek, B.**, and Macedo, J. (2025). "Challenges in NorSand for modeling stress-relief paths in tailings storage facilities." *Tailings and Mine Waste Conference 2025*.
2. **Srinivas Vivek, B.**, and Macedo, J. (2024). "Fabric effects on the onset of static liquefaction under drained conditions: Implications for mine tailings." *Tailings and Mine Waste Conference 2024*.
3. **Srinivas Vivek, B.**, Macedo, J., and Petalas, A. L. (2023). "On the effects of fabric on the instability onset under constant shear drained loading." *EMI Conference 2023*.
4. **Srinivas Vivek, B.**, Macedo, J., Petalas, A. L., and Arson, C. (2023). "Effects of initial consolidation on the triggering of static liquefaction considering fabric effects." *Geo-Congress 2023*. [PDF]
5. **Srinivas Vivek, B.**, Macedo, J., Petalas, A. L., and Arson, C. (2022). "Effects of soil fabric on the triggering of static liquefaction: Implications for mine tailings." *Tailings and Mine Waste Conference 2022*.
6. **Srinivas Vivek, B.**, Macedo, J., Petalas, A. L., and Arson, C. (2022). "On the effects of fabric on the instability surface of granular materials." *EMI Conference 2022*.
7. **Srinivas Vivek, B.**, Wang, G., Duruo, H., and Feng, J. (2019). "Fabric evolution in post-liquefaction and re-liquefaction of granular soils using 3D discrete element modeling." *Proceedings of the 7th International Conference on Earthquake Geotechnical Engineering*, Rome, Italy, vol. 4, pp. 1461–1468. CRC Press.
8. **Srinivas Vivek, B.**, and Wang, G. (2019). "DEM modelling of Toyoura sand considering real particle shapes." *23rd Annual Conference of HKSTAM and 15th Jiangsu-Hong Kong Forum on Mechanics and its Application*, Hong Kong, p. 41.
9. **Srinivas Vivek, B.**, and Mohanty, S. (2016). "1D ground response analysis of Bhubaneswar soil in India." *6th International Conference on Recent Advances in Geotechnical Earthquake Engineering and Soil Dynamics*, New Delhi, India.

Awards & Honors

- **George F. Sowers Distinguished Graduate Student Award**, Georgia Tech, 2026.
- **Outstanding Geomaterials Research Award**, Georgia Tech, 2025.
- **Paul W. Mayne Award**, Georgia Tech, 2024.
- **Best Poster Award**, Annual Geotechnical Student Poster Symposium, Geo-Institute & Georgia Tech, 2024.
- **Postgraduate Scholarship**, Hong Kong University of Science and Technology, 2017–2019.

Teaching Experience

Graduate Teaching Assistant *Jan. 2021–May 2025*
Georgia Institute of Technology

- **Courses:** Introduction to Geotechnical Engineering; Data Analytics in Civil and Environmental Engineering.
- Led data-analytics tutorials connecting statistical learning with engineering data; supported geotechnical laboratory demonstrations, office hours, and assessment.

Graduate Teaching Assistant *Feb. 2018–May 2019*
Hong Kong University of Science and Technology

Teaching Interests: Soil Mechanics; Engineering Properties of Soils; Geotechnical Earthquake Engineering; Finite Element Methods; Advanced Soil Mechanics; Constitutive Modelling; Computational Geomechanics; Site Characterization; Data Analytics for Civil Engineering.

Student Mentoring

Graduate Research Mentor, Dr. Jorge Macedo's Research Group *Aug. 2023–Present*
Georgia Institute of Technology
- Mentored a master's student in plasticity theory and NorSand implementation within a particle finite element framework for cone penetration analysis.
- Currently mentor a doctoral student extending this work by evaluating NorSand's capabilities and limitations for cone penetration modelling.

VIP Mentor, DeepMeLT and Extreme Events Engineering Groups *Sep. 2021–May 2025*
Georgia Institute of Technology
- Mentored undergraduate researchers through weekly meetings, technical discussions, and research guidance within Georgia Tech's Vertically Integrated Projects program.

Student Academic Mentor, Geosystems Engineering *Jan. 2024–Dec. 2024*
Georgia Institute of Technology
- Mentored a first-year graduate student through monthly meetings focused on academic progress, professional development, and graduate-community integration.

Student Academic Mentor *Aug. 2015–Apr. 2016*
Indian Institute of Technology Guwahati

Invited Seminars & Selected Professional Development

- **Invited expert lecture**, "Toward Performance-Based Assessment in Tailings Engineering," Indian Geotechnical Society Dhanbad chapter, IIT (ISM) Dhanbad, India, 2026.
- **Invited speaker**, "Fabric Effects in Static Liquefaction," Barr Engineering Co., 2023.
- **Invited lecture**, "NorSand and Its Visualization," Critical State Soil Mechanics courses, Colorado School of Mines and Georgia Tech.
- **Selected participant**, ALERT Olek Zienkiewicz School on Constitutive Modelling of Geomaterials, Prague, Czech Republic, 2025.
- Participant, APRU-IRIDeS Multi-Hazards Summer School, Tohoku University, Japan, 2018.

Academic & Professional Service

Peer Review Service

Ad hoc reviewer for *Journal of Geotechnical and Geoenvironmental Engineering* (ASCE), *International Journal of Geomechanics* (ASCE), and *Open Geomechanics*.

Vice President, Geotechnical Society *Aug. 2023–Aug. 2024*
Georgia Institute of Technology
- Managed organizational finances and helped organize academic and community events for the Geosystems graduate-student community.

General Secretary of Academics, Association of Civil Engineers *Oct. 2015–Oct. 2016*
Indian Institute of Technology Guwahati

Open-Source Research & Educational Resources

- **Enhancing NorSand Model Understanding Through Visualizations** - interactive educational resource. [Project]
- **DEM Visualization Toolkit** - open-source toolkit for analysis and visualization of Discrete Element Method data. [GitHub]